Singapore School Kinderland

Continual Assessment 1

2026 - 2027

Secondary 2

Computer Science

Part 2 (Answer Key)

Date: _____ Name: **TEACHER ANSWER KEY**

Instructions:

- Start answering the questions on time.
- Follow all instructions carefully.
- Answer all questions.
- Submit your paper within the given time.

Total marks	**30**
Part 2	**25**
CW	**5**
Marks Obtained	**25 / 25**

Section A:

MCQ: **(5x1=5 marks)**

For each question 1-5, four options are given. Circle the correct option i, ii, iii or iv.

1. When a large file is sent across a network, it is split into smaller units called packets. What is the typical size of each packet?

i. 1 Byte
ii. **64 KiB**
iii. 10 GiB
iv. 500 MiB

2. In packet switching, which network device reads the destination IP address and directs each packet along the best route?

i. Monitor
ii. Keyboard
iii. **Router**
iv. Scanner

3. A standard USB cable consists of a four-wire shielded cable. What are the 4 wires used for?

i. 4 wires for power only
ii. **2 wires for power and 2 wires for data**
iii. 4 wires for sound only
iv. 1 wire for power and 3 wires for video

4. What is the maximum recommended cable length for a standard USB connection before signal hub repeaters are needed?

i. **5 metres**
ii. 500 metres
iii. 5 kilometres
iv. 100 kilometres

5. Which transmission method sends data bits one after another (one bit at a time) down a single channel or wire?

i. Parallel data transmission
ii. **Serial data transmission**
iii. Skewed data transmission
iv. Analogue transmission

Section B:

Matching Terms: **(5x1=5 marks)**

6. Match each term in Column A with its correct description in Column B:

Column A (Term)	Column B (Description)	Answer
i. Payload	**A.** Marks the end of the packet and contains error-checking data (CRC).	**i → C**
ii. Packet Trailer	**B.** Bits arrive out of synchronization across parallel wires over long distances.	**ii → A**
iii. Bit Skew	**C.** The actual user data being carried inside the packet (about 64 KiB).	**iii → B**
iv. Crosstalk	**D.** Computer automatically detects a USB device and loads its driver.	**iv → E**
v. Plug and Play	**E.** Electromagnetic interference between adjacent wires in parallel cables.	**v → D**

Section C:

Direction of Data Transmission: **(3x2=6 marks)**

7. Complete the table below by stating the direction of data flow and one real-life example:

Mode	How Data Flows (1 mark each)	Real-Life Example (1 mark each)
i. Simplex	**Data travels in ONE direction only (from sender to receiver).**	**Keyboard to CPU / Computer to Printer / Television broadcast.**
ii. Half-Duplex	**Data travels in BOTH directions, but NOT at the same time (one at a time).**	**Walkie-talkie (two-way radio).**
iii. Full-Duplex	**Data travels in BOTH directions simultaneously (at the exact same time).**	**Broadband internet / Telephone call / Video call.**

Section D:

Serial vs. Parallel Transmission: **(4x1=4 marks)**

8. Tick (✓) whether each statement describes Serial or Parallel data transmission:

No.	Statement / Feature	Serial (✓)	Parallel (✓)
i.	Bits are sent one after another along a single wire or channel.	✓	
ii.	Multiple bits (e.g., 8 bits / 1 byte) are sent simultaneously across several wires.		✓
iii.	Most reliable over long distances because there is zero risk of bit skew.	✓	
iv.	Used inside a computer motherboard (internal buses) over very short distances.		✓

Section E:

Short Questions (Packet Switching & USB): **(2+3=5 marks)**

9. State any TWO items of information stored inside a Packet Header: (2 marks)

i. Sender (originating) IP address and Destination (receiver) IP address.

ii. Packet sequence number (so packets can be reassembled in the correct order) / Hop number.

10. State TWO advantages and ONE limitation of using the Universal Serial Bus (USB) interface: (3 marks)

i. Advantage 1: Devices are automatically detected and configured when plugged in (Plug and Play).

ii. Advantage 2: Connectors only fit one way (preventing incorrect connection) and USB cable supplies +5V power.

iii. Limitation: Maximum cable length is limited to about 5 metres without using powered USB hubs.